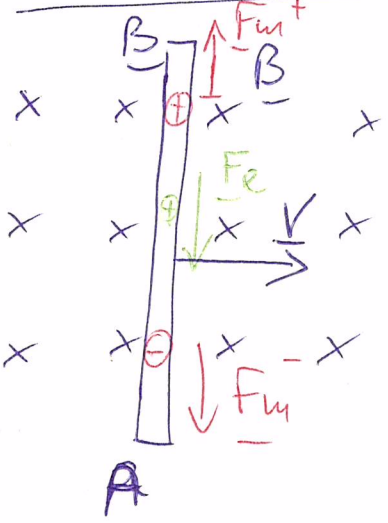
10 Electro-magnetic induction

I Motional EMI



Lorentz force:

$$\underline{F}_m = q\,\underline{v} \times \underline{B}$$

Charge separation

↳ Electrostatic force in opposing direction.

In equilibrium:

$$\underline{F}_e + \underline{F}_m = 0$$

$$q\underline{E}_{st} + q\underline{v} \times \underline{B} = 0$$

$$\Downarrow \quad \underline{E}^x : \text{external charge transporting force!}$$

$$\underline{E}^* = \underline{v} \times \underline{B} = \underline{E}_{ind}$$

↑ can consider as a new type of E

in this experiment

$$\underline{E}_{st} = \boxed{-\underline{E}_{ind}} = -\underline{v} \times \underline{B}$$

The static potential difference:

$$U^A(B)_{ST} = -\int_A^B \underline{E}_{ST}\,d\underline{r} = -\int_A^B -\underline{v} \times \underline{B}\,d\underline{r} = \int_A^B \underline{v} \times \underline{B}\,d\underline{r}$$

↑ positive!

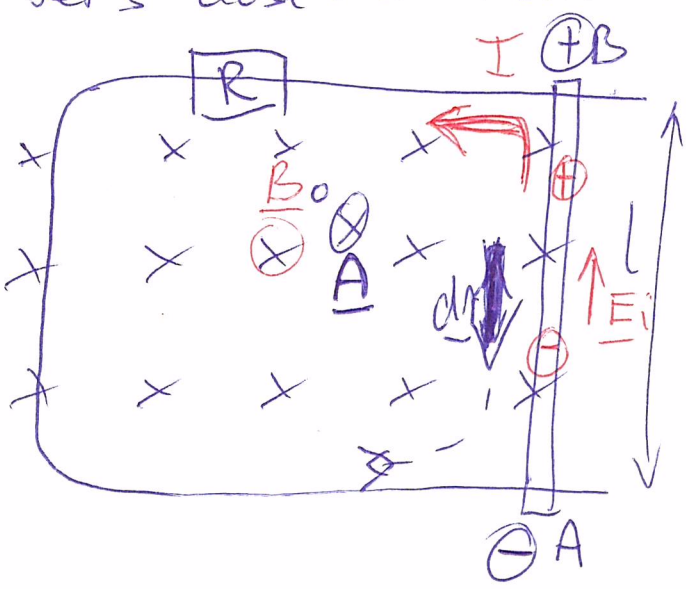
The external ~~volta~~ Electro-Motive-Force:

$$\mathcal{E} = -\int_A^B \underline{E}^*\,d\underline{r} = -\int_A^B \underline{v} \times \underline{B}\,d\underline{r} = -U^A(B)$$

$\parallel$

U_{ind} Induced voltage NOT potential difference!

Let's close the circuit:



$B \rightsquigarrow$

$$U_{ind} = -\int_A \underline{v} \times \underline{B}\, d\underline{r} = -\underline{v} \times \underline{B} \cdot \underline{l}$$

orientation of $d\underline{r}$: right hand rule with respect to B

$$U_{ind} = -\underline{v} \times \underline{B} \cdot \underline{l} = vBl = Bl \cdot \frac{\Delta x}{\Delta t} = B\frac{\Delta A}{\Delta t} = \frac{\Delta B \cdot A}{\Delta t}$$

$$= \frac{\Delta \Phi_B}{\Delta t}$$ the induced voltage is equal to the time derivative of the magnetic induction flux!

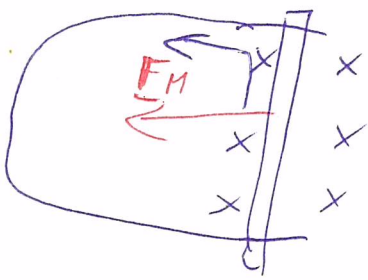
$$I_{ind} = \frac{U_{ind}}{R} = \frac{vBl}{R}$$

Current in the circuit → magnetic field is created!

the induced magnetic field (right hand rule) is an opposite direction to B_0!

↳ this reduces the magnetic field (Lenz's law)

Is there any force acting on the wire? Yes!



$$\underline{F}_M = I\, l\, \underline{u}_j \times \underline{B} = -\underline{F}_{mech}$$

to make a const velocity we have to apply an external mechanical force opposite to $\underline{F}_M$

* stg brakes the motion

The work made

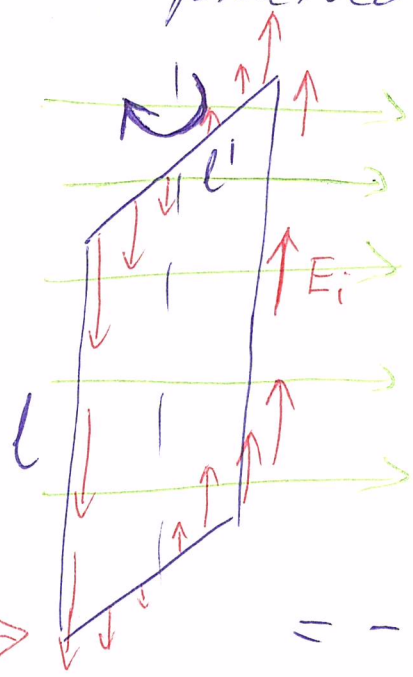
$$P_{ext} = \frac{dW_{ext}}{dt} = \underline{F}_{ext} \cdot \underline{v} = \underline{V} \cdot \underline{I}_{ind} \underline{l} (\underline{u}_j \times \underline{B}) =$$

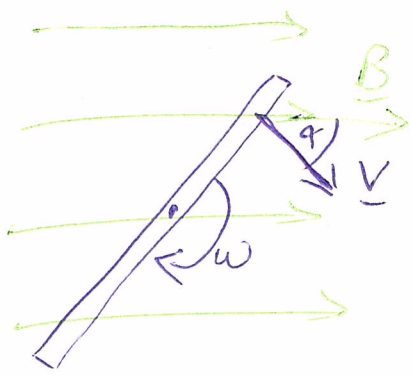
$$= V \cdot \frac{vBl}{R} \cdot l \cdot B = \underline{\underline{I_{ind}^2 \cdot R}}$$

& Mech. work transformed in Joule heat!

(Is it useful: no, but in between we have electric energy. Which is useful. We constructed an electrical power source)

More practical: rotating wire in magn field





$$\underline{E}_{ind} = \underline{v} \times \underline{B}$$

$$E_{ind} = V \cdot B \cdot \sin q$$

$$= v \cdot B \sin \omega t$$

$$U_{ind} = -\oint \underline{E}_{ind} d\underline{s} =$$

$$= -2 E \cdot l = -2 v B l \sin \omega t$$

$$U_{ind} = -2 \frac{l'}{2} \omega B \cdot l \sin \omega t = \quad \leftarrow V = r\omega = \frac{l'}{2}\omega$$

$$= -BA \sin \omega t \cdot \omega$$

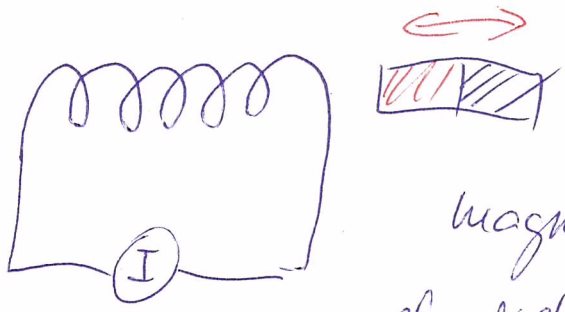
Observation: IT is NOT conservative!

$$= \frac{d\Phi_B}{dt}$$

$$(\Phi_B = B \cdot A \cdot \cos q \Rightarrow \frac{d\Phi_B}{dt} = BA - \omega \sin \omega t$$

Magnetic flux can also change without moving the wires!

experience:



Temporal change of magnetic field also creates induced el. field in a static circuit wire!

Similarly to motional induction:

$$u_{ind} (= \mathcal{E}_{ind}) = \frac{d\Phi_B}{dt} = A\frac{dB}{dt} + B\frac{dA}{dt}$$

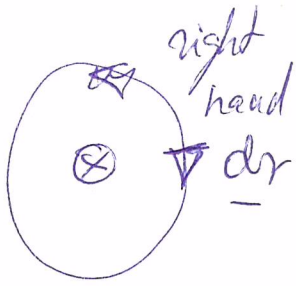
static — motional i'n

let's go back to the el. field!

$$u_{ind} = -\oint \underline{E}_{ind}\, \underline{dr} = \frac{d\Phi_B}{dt} = \frac{d}{dt}\iint \underline{B}\, \underline{ds}$$

$$\oint \underline{E}_{ind}\, \underline{dr} = -\frac{d}{dt}\iint \underline{B}\, \underline{ds}$$

$$\oint \underline{E}_{st}\, \underline{dr} = 0$$



$$\oint (\underline{E}_{st} + \underline{E}_{ind})\, \underline{dr} = \oint \underline{E}\, \underline{dr} = -\frac{d}{dt}\iint \underline{B}\, \underline{ds}$$ Faraday's law

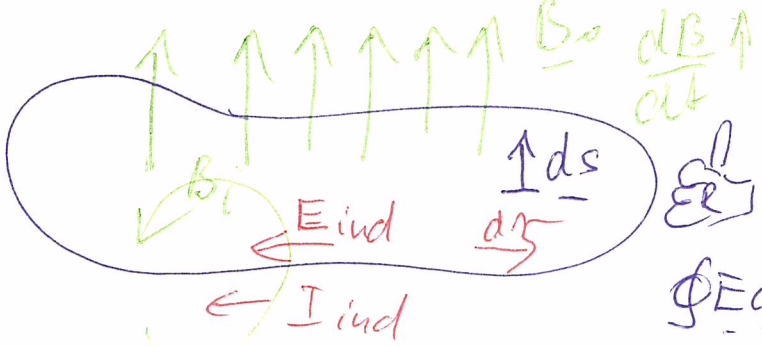
$$\oint \underline{E}\, \underline{dr} = \iint \operatorname{rot} \underline{E}\, \underline{ds} = \iint \frac{d}{dt}\underline{B}\, \underline{ds}$$ (only for static ind.)

Ch 17?

$$\operatorname{rot} \underline{E} = -\frac{\partial \underline{B}}{\partial t}$$

Induced el. field is rotational but source-less!!

Lenz's law for static circuits:



$\frac{d\underline{B}}{dt} \cdot \underline{ds} > 0$

$\oint \underline{E}\,\underline{dl} = -\iint \frac{d\underline{B}}{dt}\,\underline{ds} < 0$ negative

$\underline{E}\,\underline{dl}$ negative $\Rightarrow \underline{E} \uparrow\downarrow \underline{dl}$

$\underline{I}$ ind flows against $\underline{dl}$ $\rightarrow$ induced current

induced magnetic field due to

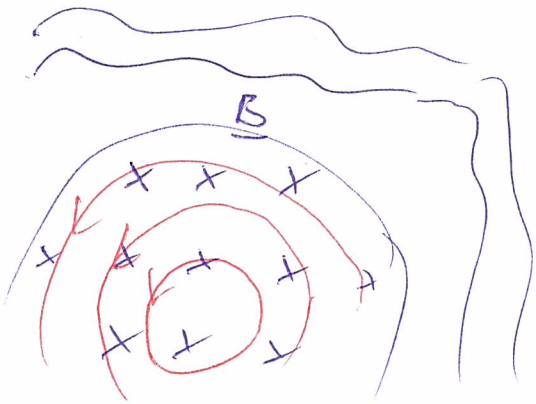
opposite to original $\underline{B}$! Lenz!

1. $B\uparrow$
2. $B\downarrow$

Important note: Just like electrostatic field, induced el. field is there, even if there is no conductor in the space! (also in vacuum!)

Eddy currents (Eddy, whirl, vortex, water twist, curly, rotation ↑ hair)

Let's investigate 2D extended conductor (plate)



suppose that $\underline{B}$ is growing! ↑ uniform.

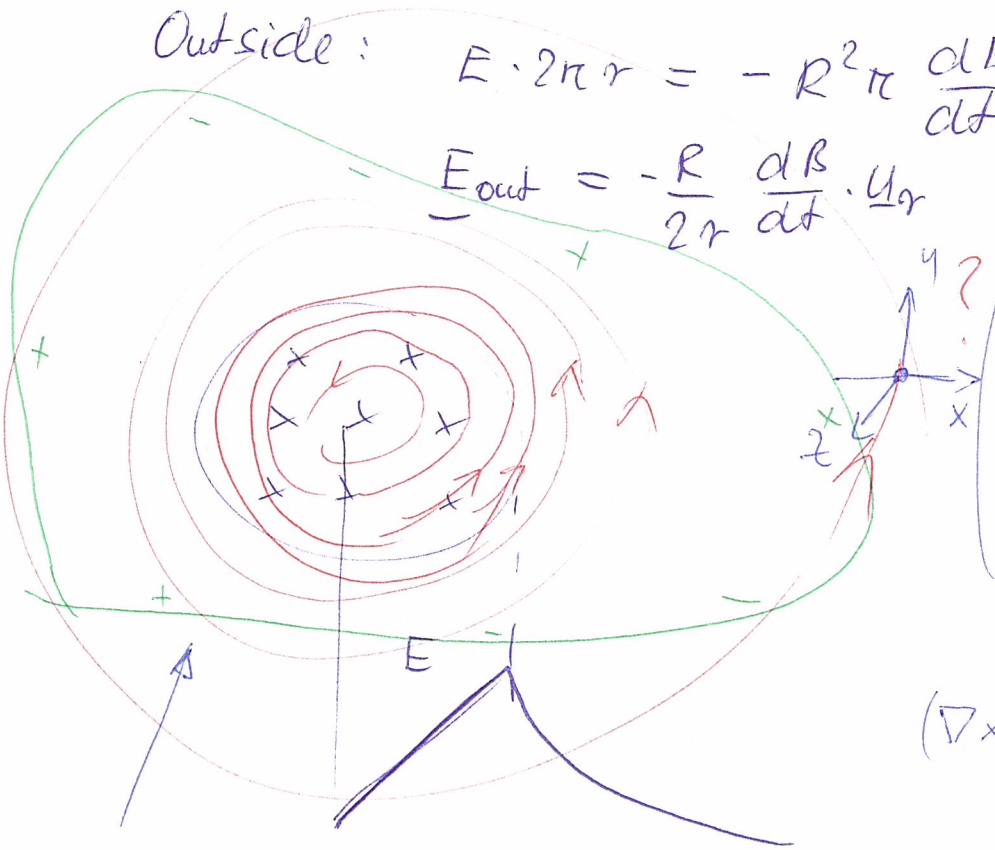
$\oint \underline{E}\,\underline{dl} = -\frac{d}{dt}\iint \underline{B}\,\underline{ds} = -r^2\pi \frac{dB}{dt}$

$\parallel$

$E \cdot 2\pi r$

$$\underline{E} = -\frac{r}{2}\frac{dB}{dt}\underline{u}_{tight}$$

current density: $\underline{j} = \sigma \underline{E}$



$$\text{rot}\, E = -\frac{\partial B}{\partial t} = 0 \quad \text{outside}$$

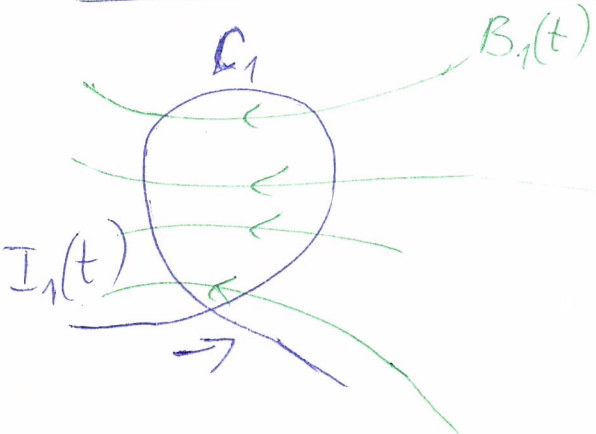
$$E = \begin{pmatrix} 0 \\ E_y \\ 0 \end{pmatrix} \qquad \partial_x E_y - \partial_z E_y$$

$$\nabla \times E = \begin{pmatrix} -\partial_z E_y \\ 0 \\ \partial_x E_y - \partial_y E_x \end{pmatrix}$$

$$(\nabla \times A)_t = \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\varphi) - \frac{\partial A_r}{\partial \varphi}\right]$$

What happens if the plate is finite? Charge accumulation!

Mutual inductance and self-inductance



A part of the magn. flux goes through $C_2 \rightarrow$ EMI

$$u_{ind_2} = \frac{d\Phi_{B_2}}{dt} \qquad \Phi_{B_2} \sim I_1$$

$$\Phi_{B_2} = M_{21} I_1$$

$$u_{i2} = M_{21}\frac{dI_1}{dt}$$

$$[M_{21}] \perp Vs = 1H$$

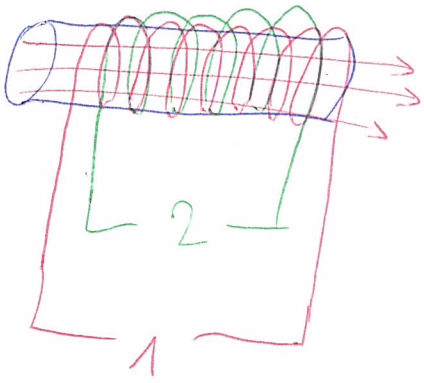
Joseph Henry

CH 18: $M_{12} = M_{21}$ Prove!

Minimize mutual induction: go far away

Maximiz -11-1 : All flux of C_1 goes through C_2

E.G. 2 solenoids overlapping on a cylinder



Field of coil 1:

$$B_1 = \frac{\mu_0 N_1 I_1}{l}$$

Induction flux in the 2nd coil

$$\Phi_{B_2} = N_2 \cdot A \cdot B_1 = \underbrace{\frac{\mu_0 N_1 N_2 A}{l} I_1}_{M\,(=M_{12}=M_{21})}$$

Maximize even more: $\mu_0 \to \mu_0 \cdot \mu_r$

$$U_{ind_2} = \frac{d\Phi_{B_2}}{dt} = M \frac{dI_2}{dt}$$

Self induction



← Lenz OK! (induction is against the changes creating it!)

Shall we calculate for a solenoid?

No. See up ↑ & $B(t) = \frac{\mu_0 \cdot N\, I(t)}{l}$

$$U_{ind} = \frac{d\Phi_B}{dt} = N \cdot A \cdot \mu_0 \frac{N\, I(t)}{l} = \underbrace{\frac{\mu_0 N^2 A}{l}}_{L} \frac{dI}{dt}$$

(Unipolar induction, Electromagnetic cannon?)